

Méthode Adjoin

$$C = \begin{bmatrix} -1 & 2 & 5 \\ 4 & -3 & 1 \\ 0 & 2 & 3 \end{bmatrix} \begin{matrix} -1 & 2 \\ 4 & -3 \\ 0 & 2 \end{matrix} \begin{matrix} (9+0+40) - (24+(-2)+0) \\ = (49) - (22) \\ = 49 - 22 = 27 \end{matrix}$$

$$M_{41} = \begin{bmatrix} -3 & 1 \\ 2 & 3 \end{bmatrix} \quad M_{12} = \begin{bmatrix} 4 & 1 \\ 0 & 3 \end{bmatrix} \quad M_{13} = \begin{bmatrix} 4 & -3 \\ 0 & 2 \end{bmatrix}$$

$$\begin{matrix} = -9 - 2 \\ = -11 \end{matrix} \quad \begin{matrix} = 12 - 0 \\ = 12 \end{matrix} \quad \begin{matrix} = 8 - 0 \\ = 8 \end{matrix}$$

$$M_{21} = \begin{bmatrix} 2 & 5 \\ 2 & 3 \end{bmatrix} \quad M_{22} = \begin{bmatrix} -1 & 5 \\ 0 & 3 \end{bmatrix} \quad M_{23} = \begin{bmatrix} -1 & 2 \\ 0 & 2 \end{bmatrix}$$

$$\begin{matrix} = 6 - 10 \\ = -4 \end{matrix} \quad \begin{matrix} = -3 - 0 \\ = -3 \end{matrix} \quad \begin{matrix} = -2 - 0 \\ = -2 \end{matrix}$$

$$M_{31} = \begin{bmatrix} 2 & 5 \\ -3 & 1 \end{bmatrix} \quad M_{32} = \begin{bmatrix} -1 & 5 \\ 4 & 1 \end{bmatrix} \quad M_{33} = \begin{bmatrix} -1 & 2 \\ 4 & -3 \end{bmatrix}$$

$$\begin{matrix} = 2 - (-15) \\ = 2 + 15 = 17 \end{matrix} \quad \begin{matrix} = -1 - 20 \\ = -21 \end{matrix} \quad \begin{matrix} = 3 - 8 \\ = -5 \end{matrix}$$

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$$C = \begin{bmatrix} -11 & 12 & 8 \\ -4 & -3 & -1 \\ 17 & -21 & -5 \end{bmatrix}$$

$$C = \begin{bmatrix} -11 & -12 & 8 \\ 4 & -3 & 2 \\ 17 & 21 & -5 \end{bmatrix}$$

Matrix Adjoin C

$$A C^{-1} = \frac{1}{\det(C)} \operatorname{adj}(C) = \frac{1}{27} \begin{bmatrix} -11 & -12 & 8 \\ 4 & -3 & 2 \\ 17 & 21 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{-11}{27} & \frac{-12}{27} & \frac{8}{27} \\ \frac{4}{27} & \frac{-1}{9} & \frac{2}{27} \\ \frac{17}{27} & \frac{21}{27} & \frac{-5}{27} \end{bmatrix}$$

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$$C = \begin{bmatrix} -11 & 4 & 17 \\ 27 & 27 & 27 \\ -12 & -1 & 7 \\ 27 & 9 & 9 \\ 8 & 2 & -5 \\ 27 & 27 & 27 \end{bmatrix}$$

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$$C = \begin{bmatrix} -1 & 2 & 5 \\ 4 & -3 & 1 \\ 0 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\det AC = 27$$

$$C \cdot H_1(-1) = \begin{bmatrix} 1 & -2 & -5 \\ 4 & -3 & 1 \\ 0 & 2 & 3 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$H_{21} \begin{pmatrix} -4 \end{pmatrix} = \left[\begin{array}{ccc|ccc} 1 & -2 & -5 & -1 & 0 & 0 \\ 4 & -3 & 1 & 0 & 1 & 0 \\ 0 & 2 & 3 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & -2 & -5 & -1 & 0 & 0 \\ 0 & 5 & -21 & 4 & 1 & 0 \\ 0 & 2 & 3 & 0 & 0 & 1 \end{array} \right]$$

$$H_{13}(6) = \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 4 & 0 & 1 \\ 0 & 5 & 21 & 4 & 1 & 0 \\ 0 & 2 & 3 & 0 & 0 & 0 \end{array} \right]$$

$$H_{23}(-2) = \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & -1 & 0 & 1 \\ 0 & 1 & 15 & 4 & 1 & -2 \\ 0 & 2 & 3 & 0 & 0 & 1 \end{array} \right]$$

$$H_{12}(-2) = \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & -1 & 0 & 1 \\ 0 & 1 & 15 & 4 & 1 & -2 \\ 0 & 2 & 3 & 0 & 0 & 1 \end{array} \right]$$

$$H_{13} \begin{pmatrix} -2/27 \end{pmatrix} = \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & -1 & 0 & 1 \\ 0 & 1 & 15 & 4 & -1 & -2 \\ 0 & 0 & -27 & -8 & -2 & 5 \end{array} \right]$$

$$H_{23} \begin{pmatrix} 15/27 \end{pmatrix} = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -11/27 & 4/27 & 12/27 \\ 0 & 1 & 0 & -4/9 & -1/9 & 7/9 \\ 0 & 0 & 1 & -8 & -2 & 5 \end{array} \right]$$

$$H_{12} \begin{pmatrix} -1/27 \end{pmatrix} = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -4/27 & 4/27 & 11/27 \\ 0 & 1 & 0 & -4/9 & -1/9 & 7/9 \\ 0 & 0 & 1 & 8/27 & 2/27 & 5/27 \end{array} \right]$$

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$$A(s) = \begin{bmatrix} -1 & 2 & 5 \\ 4 & -3 & 1 \\ 0 & 2 & 3 \end{bmatrix}$$

$$\det C = 27$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$K_u(s)$$

$$\begin{bmatrix} 1 & 2 & 65 \\ -4 & -3 & 1 \\ 0 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$K_{21}(s)$$

$$\begin{bmatrix} 1 & 0 & 5 \\ -4 & 5 & 1 \\ 0 & -2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$K_{12}(s)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -4 & 5 & 21 \\ 0 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 2 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$K_{22}(s)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 21 \\ 0 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 3/5 & 2 & 5 \\ 4/5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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 $K_{11} (-1)$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 21 \\ 8/5 & 10/17 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 3/5 & 2/17 & 5 \\ 4/9 & 1 & 0 \\ 0 & -4/27 & 1 \end{bmatrix}$$

 $K_{13} (0/17)$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 8/5 & 4/9 & -2/7 \end{bmatrix}$$

$$\begin{bmatrix} 3/5 & 2/17 & -1/7 \\ 4/5 & 1 & -21 \\ 0 & -4/17 & 5 \end{bmatrix}$$

 K_{23} $10/109$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 10/17 & -2/7 \end{bmatrix}$$

$$\begin{bmatrix} -4/17 & 2/7 & -1/7 \\ -4/9 & 1 & -21 \\ 8/17 & -4/17 & 5 \end{bmatrix}$$

 $K_2 (-1/9)$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2/7 \end{bmatrix}$$

$$\begin{bmatrix} -4/17 & 4/17 & -1/7 \\ -4/9 & -1/9 & -1 \\ 0/17 & 2/17 & 5 \end{bmatrix}$$

 R_{23}

$$K_2: \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -4/17 & 4/17 & 1/17 \\ -4/9 & -1/9 & 1/9 \\ 0/17 & 2/17 & -5/17 \end{bmatrix}$$

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